

20.08.2023

MATHEMATICS

1. In $\triangle ABC$ and $\triangle DCB$
 $\triangle ABC \sim \triangle DCB$ (given)
 $\frac{AB}{DC} = \frac{AC}{DB}$ (corresponding sides of similar triangles).

$$AB \times DB = \underline{DC \times AC}$$

$$\therefore AB \times DB = DC \times AC$$

2. Origin = $(0, 0)$
 point A = (a, b)

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(a-0)^2 + (b-0)^2}$$

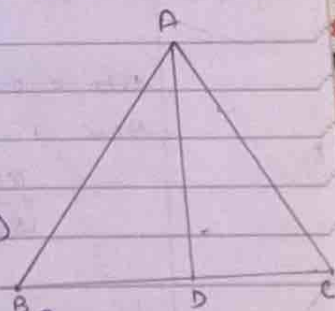
$$= \underline{\sqrt{a^2 + b^2}}$$

The distance from origin is $\sqrt{a^2 + b^2}$

3. In $\triangle ABC$ and $\triangle ADC$,
 $\angle BAC = \angle ADC$ (given)
 $\angle ACB = \angle ACD$ (common)
 $\therefore \triangle ABC \sim \triangle DAC$ (AA similarity)

$$\frac{AC}{DC} = \frac{BC}{AC} \quad \left[\begin{array}{l} \text{corresponding sides} \\ \text{of similar triangles} \end{array} \right]$$

$$\frac{AC}{BC} = \frac{DC}{AC}$$



let,

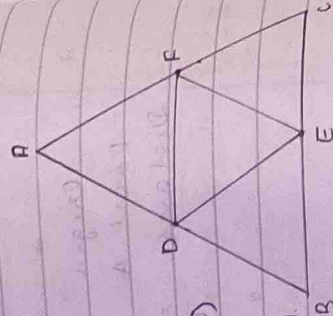
4. A = (x_1, y_1)
 B = (x_2, y_2)

$P = (x, y)$
and m_1, m_2 be the ratio
section Formula

$$(x, y) = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

$$\text{and } y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$



5. In $\triangle ABC$ and $\triangle DEF$, $\triangle ADE$,
 $\angle A = \angle A$ (common)
 $\frac{AD}{AB} = \frac{AE}{AC} = \frac{1}{2}$ & (mid point)

$\triangle ABC \sim \triangle ADE$ (SAS similarity)

$$\therefore \frac{DE}{BC} = \frac{AD}{AB} = \frac{AE}{AC} = \frac{1}{2}$$

$$\therefore \frac{DE}{BC} = \frac{1}{2}$$

We know that,

Now,

$$\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle DEF)} = \left(\frac{BC}{DE} \right)^2$$

$$\frac{\text{Area}(\triangle DEF)}{\text{Area}(\triangle ABC)} = \left(\frac{DE}{BC} \right)^2$$

$$= \left(\frac{1}{2} \right)^2$$

$$= \frac{1}{4}$$

Hence the ratio of area of ΔEFD to $\Delta ABC = 1:4$

6. ~~but~~ $A = (-3, 2)$ let the ratio be
 $B = (6, 1)$ $m_1 : m_2$
 $P = (0, y)$

Using section formula

$$(x, y) = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

$$(0, y) = \left(\frac{m_1(6) + m_2(-3)}{m_1 + m_2}, \frac{m_1(1) + m_2(2)}{m_1 + m_2} \right)$$

$$0 = \frac{6m_1 - 3m_2}{m_1 + m_2}$$

$$0 = 6m_1 - 3m_2$$

$$3m_2 = 6m_1$$

$$\frac{3}{6} = \frac{m_1}{m_2} = \frac{1}{2}$$

The ratio by which the line segment is divided is 1:2.

7. In the figure

$DE \parallel BC$

By BPT

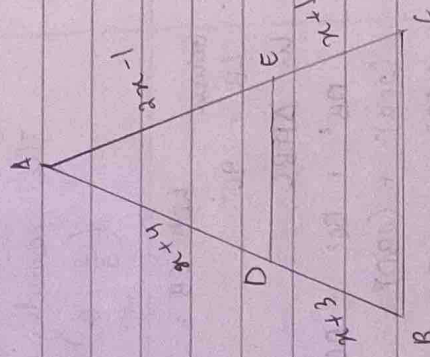
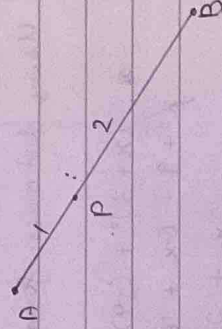
$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$\frac{x+4}{x+3} = \frac{2x-1}{x+1}$$

$$(x+4)(x+1) = (2x-1)(x+3)$$

$$x^2 + x + 4x + 4 = 2x^2 + 6x - x - 3$$

$$x^2 + 5x + 4 = 2x^2 + 5x - 3$$



$$x^2 + 4 = 2x^2 - 3$$

$$4 + 3 = 2x^2 - x^2$$

$$7 = x^2$$

$$\therefore x = \pm\sqrt{7}$$

The value of $x = \pm\sqrt{7}$

8. The points are $A(3, 4)$ and $B(2, 5)$
let the point on the x -axis be $P(x, 0)$

Now,

$$AP^2 = BP^2$$

$$AP^2 = BP^2$$

Using distance formula,

$$(x+3)^2 + (-4)^2 = (x-2)^2 + (-5)^2$$

$$x^2 + 9 + 6x + 16 = x^2 + 4 - 4x + 25$$

$$6x + 25 = 4 - 4x + 25$$

$$6x = 4 - 4x$$

$$6x + 4x = 10x = 4$$

$$x = \frac{4}{10} = \frac{2}{5}$$

The coordinates of the point on x -axis is $\left(\frac{2}{5}, 0\right)$

Given:

$$\angle B = 90^\circ$$

In $\triangle ABC$,

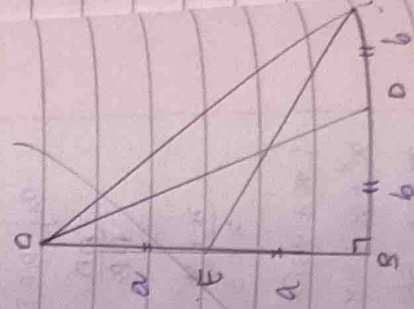
$$AB^2 + BC^2 = AC^2 \quad (\text{Pythagoras Theorem})$$

$$(2EB)^2 + (2BD)^2 = 25$$

$$4EB^2 + 4BD^2 = 25$$

$$EB^2 + BD^2 = \frac{25}{4}$$

$$a^2 + b^2 = \frac{25}{4}$$



9. construction : Join ED.

Let $AE = EB = a$

and $BD = DC = b$

By Pythagoras theorem,

$$AC^2 = (2a)^2 + (2b)^2 = 4a^2 + 4b^2$$

$$ED^2 = a^2 + b^2$$

$$AD^2 = (2a)^2 + b^2 = 4a^2 + b^2$$

$$CE^2 = a^2 + (2b)^2 = a^2 + 4b^2$$

Now,

$$AD^2 + CE^2 = 4a^2 + b^2 + a^2 + 4b^2$$

$$= 5a^2 + 5b^2$$

$$ED^2 + AC^2 = 4a^2 + 4b^2 + a^2 + b^2$$

$$= 5a^2 + 5b^2$$

$$\therefore AD^2 + CE^2 = ED^2 + AC^2$$

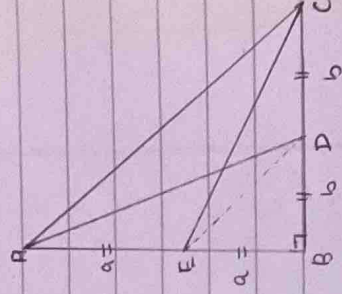
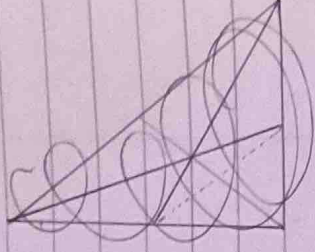
$$\left(\frac{3\sqrt{5}}{2}\right)^2 + CE^2 = \frac{25}{4} + 25$$

$$\therefore CE^2 = \frac{25}{4} + 25 - \frac{45}{4}$$

$$= \frac{105 - 45}{4} = \frac{80}{4}$$

$$CE^2 = 20$$

$$\therefore CE = \sqrt{20} = 2\sqrt{5} \text{ cm}$$



In $\triangle ABD$,

$$AB^2 + BD^2 = AD^2 \quad (\text{Pythagorean Theorem})$$

$$(2EB)^2 + (AD)^2 = \left(\frac{3\sqrt{5}}{2}\right)^2$$

$$4EB^2 + BD^2 = \frac{45}{2}$$

$$4a^2 + b^2 = \frac{45}{2}$$

10. Let the points be

$$A = (6, 8)$$

$$B = (3, 9)$$

$$C = (-2, -2)$$

$$\text{and } D = (1, -9)$$

$$AB = \sqrt{(6-3)^2 + (8-9)^2} = \sqrt{9+1} = \sqrt{10}$$

$$BC = \sqrt{(3-2)^2 + (9-7)^2} = \sqrt{25+81} = \sqrt{106}$$

$$CD = \sqrt{(-2-1)^2 + (-2+9)^2} = \sqrt{9+1} = \sqrt{10}$$

$$AD = \sqrt{(6-1)^2 + (8+9)^2} = \sqrt{25+81} = \sqrt{106}$$

Now,

$$AC = \sqrt{(6+2)^2 + (8+2)^2} = \sqrt{64+100} = \sqrt{164}$$

~~$$BD = \sqrt{(3-1)^2 + (9-9)^2} = \sqrt{4+0} = \sqrt{4} = 2$$~~

8

$$BD = \sqrt{(3-1)^2 + (9+9)^2} = \sqrt{4+64} = \sqrt{68}$$

Now since, the opposite sides are equal but the diagonals are not equal.
The quadrilateral formed is a parallelogram.